

Original summary

After **data import**, you will be shown a summary of per spot/cell gene count metrics. This summary presents the number of counts and genes per spot (or cell, depending on the spatial transcriptomics technology), for each sample. Here a gene is counted if at least one transcript was detected for that gene in at least one spot/cell. This summary can be downloaded to a comma-delimited file if necessary (red arrow). This tutorial will start by clicking the **Filter data** tab.

spatialGE

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- Phenotyping
- Differential expression

QC & Data Transformation
Documentation

- Original summary:** A table displaying the minimum, average, and maximum number of counts per ROI/spot/cell, as well as the minimum, average, and maximum number of genes per ROI/spot/cell. Genes count as expressed if at least one count was detected in at least one ROI/spot/cell.
- Filter data:** Remove ROIs/spots/cells or genes from the data set. The filter can be executed on all samples or a subset. Internally, spatialGE executes the filter in the following order: 1. Samples specified in "Select samples to apply this filter", 2. Genes specified by name in "Filter genes", 3. ROIs/spots/cells and/or genes based on min and max counts.
- Normalize data:** Transform gene expression counts, with log-normal or SCTransform as options. The normalization procedure does not require filtering, but removal of low-count spots/cells or low count genes is recommended.
- Pseudobulk analysis:** Explore sample-level gene expression. Pseudobulk analysis combines gene counts across all spots/cells and produce pseudo-RNAseq samples. This analysis allows exploration of sample-level transcriptional patterns. Does not require execution of the Filter Data or Normalize Data sections.
- Quilt plot:** Generates spatial (quilt) plots showing spot/cell-level metadata. Users can assess QC metrics such as gene counts per spot or total counts per spot and detect potential technical artifacts.

Original summary | **Filter data** | Normalize data | Pseudo-bulk analysis | Quilt plot

Search...

Sample name	Total spots/cells	Total genes	Min. counts	Avg. counts	Max. counts	Min. genes	Avg. genes	Max. genes
ductal_carcinoma	2518	17651	253	13320	43555	225	4776	8414
breast_cancer_section1	3813	22968	524	18607	72315	393	5064	9594
breast_cancer_section2	4015	23155	96	18425	80748	81	5004	10020
breast_cancer_wta	4325	23812	35	11751	59080	34	3635	9059

[Download summary](#)

Filter data

In the **Filter data** tab, please click the “Select samples to apply this filter” text. A table similar to **Original Summary** is presented. You can select individual samples that *you do not wish to apply the filter* (red arrows). In this tutorial, however, no samples will be excluded from the filter as most users will probably want to apply filters to all samples in a project.

Please click in the next section **Filter spots/cells**.

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Documentation

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Original summary | **Filter data** | Normalize data | Pseudo-bulk analysis | Quilt plot

Select samples to apply this filter

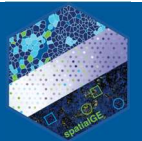
CONTINUOUS SCROLLING

Search...

☒	Sample name	Total spots/...	Total genes	Min. counts	Avg. counts	Max. counts	Min. genes	Avg. genes	Max. genes
☒	ductal_carc...	2518	17651	253	13320	43555	225	4776	8414
☒	breast_can...	3813	22968	524	18607	72315	393	5064	9594
☒	breast_can...	4015	23155	96	18425	80748	81	5004	10020
☒	breast_can...	4325	23812	35	11751	59080	34	3635	9059

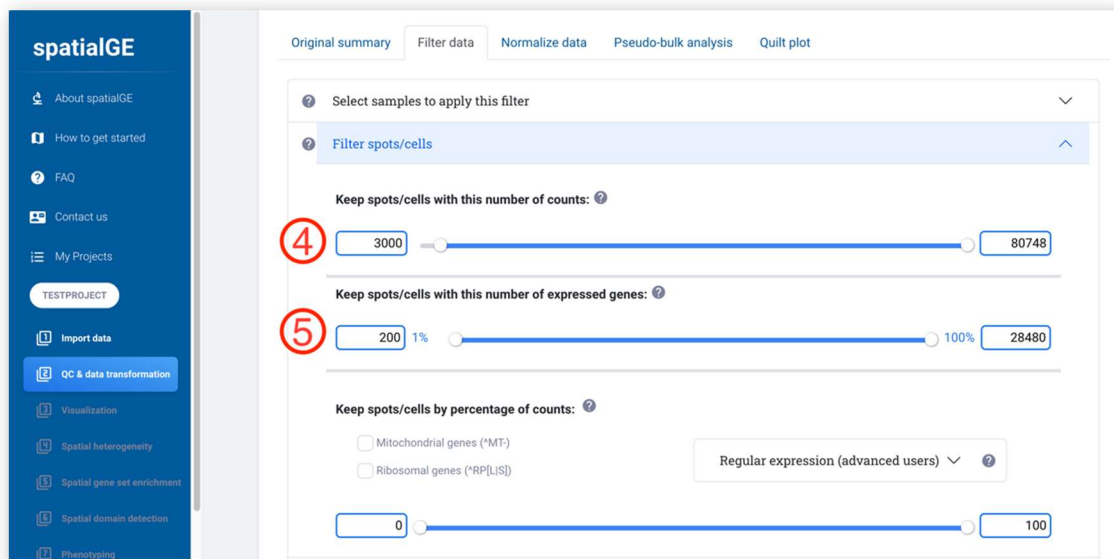
Filter spots/cells

Filter genes

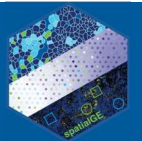


The **Filter spots/cells** section removes spots or cells based on the number of counts. The first filter in this section, **Keep spots/cells with this number of counts**, defines the minimum and maximum gene counts an spot/cell should have to be kept in the data set after filtering. The user is encouraged to think about reasonable values for this filter. Since each project is different, and factors such as quality of the sample, sequencing depth, and level of necrosis, affect the number of counts per spot/cell, establishing default values is difficult. From the **Original summary** tab, it can be seen that the spot each Visium sample has spots with gene counts as low as 35. For the purpose of this example, spots with less than 3000 counts per spot will be removed, which should be enough to remove low-quality spots. For that, the user can either type in the number in the left box or move the left slider button to the right until 3000 shows in the left box. The maximum number of counts per spots will be left untouched. Nevertheless, for technologies that use image segmentation to define cells, the user might want to limit the maximum number of counts per cell to remove poorly segmented cells (“doublets”).

Next, the **Keep spots/cells with this number of expressed genes** filter removes spots or cells based on the number of expressed genes. Here, a gene is considered expressed if at least one transcript was detected in at least one spot. In general, spots with low number of expressed genes are also considered low-quality. From the **Original summary** tab, it can be seen that the sample “breast_cancer_wta” has spots with as low as 34 genes. To remove those spots, please write in the left box the number 200. Similarly, the maximum number of genes will be left untouched, but the same previous considerations apply for image segmentation-based spatial transcriptomics.



The spots can also be filtered based on the percentage of mitochondrial or ribosomal counts. Under **Keep spots/cells by percentage of counts**, the user can specify the minimum and maximum percentage of counts from mitochondrial or ribosomal genes. Here, spots will be filtered based only on mitochondrial gene counts by checking the “**Mitochondrial genes (^MT-)**” box. Next, move the *slider button on the right to the 20%* mark or type 20 in the right box. Keep the left slider button in zero. With this filter, spots that contain 0 to 20% of mitochondrial counts are kept. A 20% cutoff for mitochondrial counts is commonly used in many single-cell



studies. Here, this filter aims to remove spots from areas with cellular stress or death which may have higher expression of mitochondrial genes.

Note: The “^MT-” in the **Mitochondrial genes** check box indicates that spatialGE searches for genes with names starting by “MT” followed by a hyphen in the gene names. This filter works in Visium as mitochondrial genes are named that way; however other technologies might not follow that naming pattern. In those cases, the user can input a regular expression or token that captures the names of the genes to be used in the filter in the **Regular expression** box (green arrow).

Now, please click the **Filter genes** section.

The screenshot shows the 'Filter spots/cells' section of the spatialGE interface. The sidebar on the left contains navigation links: 'About spatialGE', 'How to get started', 'FAQ', 'Contact us', 'My Projects', 'TESTPROJECT', 'Import data', 'QC & data transformation' (highlighted), 'Visualization', 'Spatial heterogeneity', 'Spatial gene set enrichment', 'Spatial domain detection', and 'Phenotyping'. The main panel has a 'Filter spots/cells' header. Below it, there are three filter sections: 'Keep spots/cells with this number of counts' (slider from 3000 to 80748), 'Keep spots/cells with this number of expressed genes' (slider from 200 to 28480), and 'Keep spots/cells by percentage of counts' (checkboxes for 'Mitochondrial genes (*MT-)' and 'Ribosomal genes (*RP[LIS])', and a 'Regular expression (advanced users)' dropdown with a green arrow pointing to it). Below these is a slider from 0 to 20. At the bottom, there is a 'Filter genes' section (circled 8) and an 'APPLY FILTER' button.

The **Filter genes** section removes genes based on the total number of counts or spots/cells where those genes are expressed. The user can also remove genes by name using the two boxes at the top of the section. This functionality is especially useful when removing negative probes or housekeeping genes used for background noise correction (e.g., CosMx). The user can type in the left text box (“Search genes here...”), which will fill the box under it with genes that match the typed text. By clicking on a gene name, it is passed to the box on the right (“Excluded genes”). If the user decides to remove all mitochondrial or ribosomal genes from the sample, it can be accomplished by clicking one or the two corresponding check boxes (**Remove mitochondrial genes** and/or **Remove ribosomal genes**). Similarly, you can search for regular expression that matches genes to be removed (“Remove genes using regular expression”).

In this example, genes will not be removed by name. Instead, the two sliders will be used. The **Keep genes with counts between** slider specifies the minimum and maximum number of counts across all spots for a gene to be kept. Similar to the filters in the previous sections, the values used in the gene filters are relative to each experiment, and hence defaults or guidelines are difficult to establish. For this example, please type 2000 in the



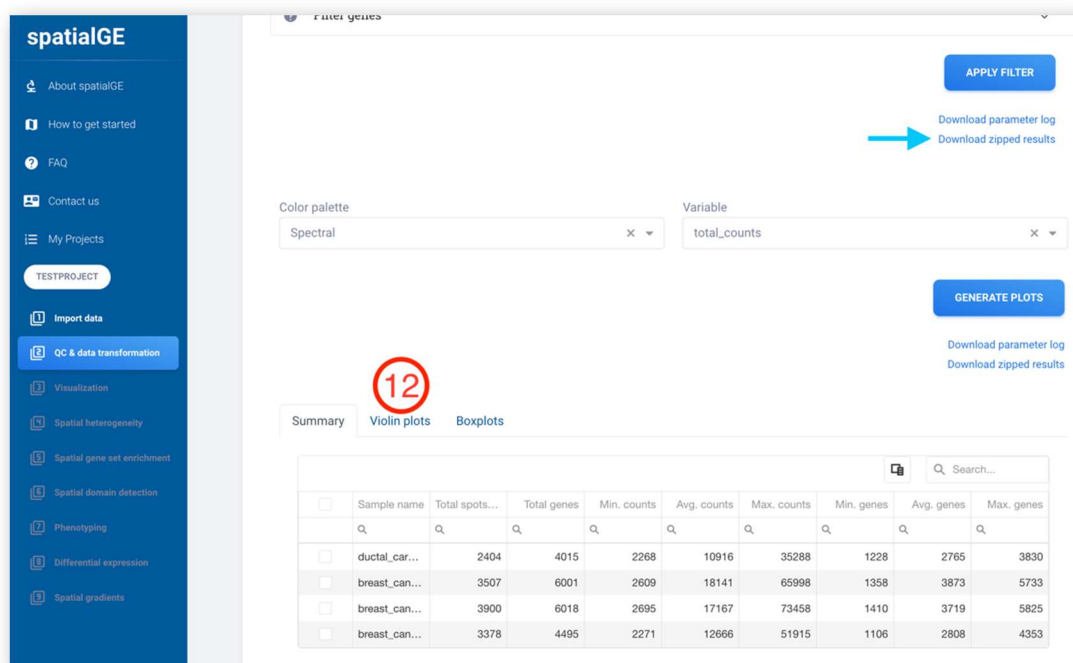
left box, which means that a gene must have at least 2000 counts across the sample (not in each spot) to be kept. The following filter is **Keep genes expressed in**, which controls the number of spots/cells a gene must be expressed in to be kept (i.e., at least one count in a spot/cell). Before using this filter, the user is encouraged to think about the particularities of their samples. For example, if a gene is not expressed in a spot, it does not necessarily mean that the technology failed to provide data, given that the same gene can be expressed abundantly in other spots. A reasonable value would be to filter genes expressed in at least 20 spots. The expectancy here is that lowly expressed genes could be removed to avoid problems with statistical testing (e.g., differential expression). Please, type the number 20 in the left box of this filter.

By clicking the **APPLY FILTER** button, the filters selected in all the previous section is executed.

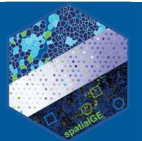
The screenshot shows the 'Filter genes' interface in spatialGE. The sidebar on the left contains navigation links: About spatialGE, How to get started, FAQ, Contact us, My Projects, TESTPROJECT, Import data, **QC & data transformation** (highlighted), Visualization, Spatial heterogeneity, Spatial gene set enrichment, Spatial domain detection, and Phenotyping. The main panel has a 'Filter genes' header. Below it are search and exclusion options: 'Search genes here...' and 'Excluded genes:'. There are checkboxes for 'Remove mitochondrial genes (*MT)' and 'Remove ribosomal genes (*RP[LS])'. A dropdown menu shows 'Remove genes using regular expression (advanced users)'. Two sliders are present: 'Keep genes with counts between:' with a value of 2000 (highlighted by red circle 9) and a range up to 2850596; and 'Keep genes expressed in:' with a value of 20 (highlighted by red circle 10) and a range up to 4325. The 'APPLY FILTER' button is highlighted by red circle 11.

A pop-up will appear asking if an email notification should be sent once the filter is applied. Then, a new summary of genes and counts is presented at the bottom of the page. This summary shows the results of the filtering procedure. Users can compare these numbers with the **Original summary** tab to assess the effects of the filtering procedure on the data. The minimum number of counts and genes per spot seem to be in the same approximate range, which is result of a filtering procedure focused on the minimum values and aiming to remove low-quality spots. Notice this summary can also be downloaded as a .csv file using the **Download zipped results** link (light blue arrow) under the **APPLY FILTER** button. The same link can be used to download a file with all the parameters used for filtering, facilitating replication and reporting of analyses.

Graphical summaries are available by clicking the **Violin plots** or **Boxplots** tab.



In the **Violin plots** tab, an auto-generated plot shows the distribution of counts per sample. Each point within the violins represents a spot/cell. With this visual summary is easy to notice that the “breast_cancer_section1”, “breast_cancer_section2”, and “breast_cancer_wta” samples have higher counts compared to “ductal_carcinoma”. This is expected as the “ductal_carcinoma” data comes from a formalin-fixed paraffin-embedded (FFPE) tissue. The current version of spatialGE allows to plot a second QC metric. By clicking on the **Variable** drop-down, the “total_genes” metric can be selected. If a different color palette is desired, you can select it in the **Color palette** drop-down. The palettes have been sourced from the [Khroma](#) and [RColorBrewer](#) R packages. By clicking the **GENERATE PLOTS** button, and after that return to the **Violin plots** tab, the number of genes per spot are presented. Notice that these plots can be downloaded using the **PDF/PNG/SVG** buttons below the plots.

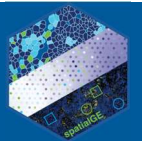


Normalize data

The next step in the analysis is to click on the **Normalize data** tab. The normalization of the data is required to enable additional analysis modules in spatialGE. Currently, two normalization methods are available: library-size normalization followed by log-transformation (**Log-normal**) and the variance stabilization method **SCTransform**. The latter uses negative binomial regression residuals to reduce the effect of spot-to-spot total count variation, with the expectation of extracting more biologically meaningful differences. For further details, please refer to the [technical article](#) describing the method.

Here, the user selects the method to be used by clicking on the respective radio button. In this example, the default **Log-normal** will be used. Notice that the parameter **Scaling factor** is available for the **Log-normal** transformation. This parameter must not be confounded with the JSON file used in the [data import](#) process. Instead, this **Scaling factor** is a rough way to scale spot/cell counts to a certain value during normalization. The default (10,000) results from the expectation that most spots have that number of counts. The user can intuit that this factor must be adjusted according to the technology used. For example, in single-cell level spatial transcriptomics, this parameter will probably need to be set to a lower value.

To perform the data normalization procedure, click the **NORMALIZE DATA** button.



low-count spots/cells or low count genes is recommended.

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- Quilt plot:** Generates spatial (quilt) plots showing spot/cell-level metadata. Users can assess QC metrics such as gene counts per spot or total counts per spot and detect potential technical artifacts.

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Select a method:

Log-normal

Log-normalization: Counts are normalized to library size (total number of counts per spot or cell), followed by multiplication with a scaling factor. Then, scaled counts are log-transformed (natural logarithm + 1).

Scaling factor:

☒ Use Log-normal

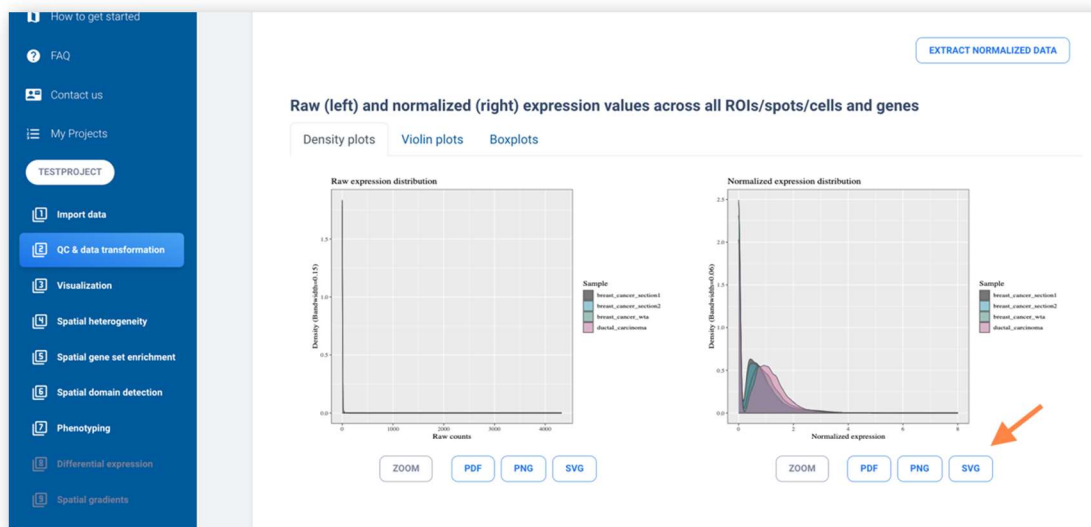
SCTransform (Seurat implementation)

SCTransform: Method implemented in Seurat for count transformation and reduction of technical effects. The method applies regularized negative binomial regression with counts per spot or cell as covariate (See their [technical article](#) for more details).

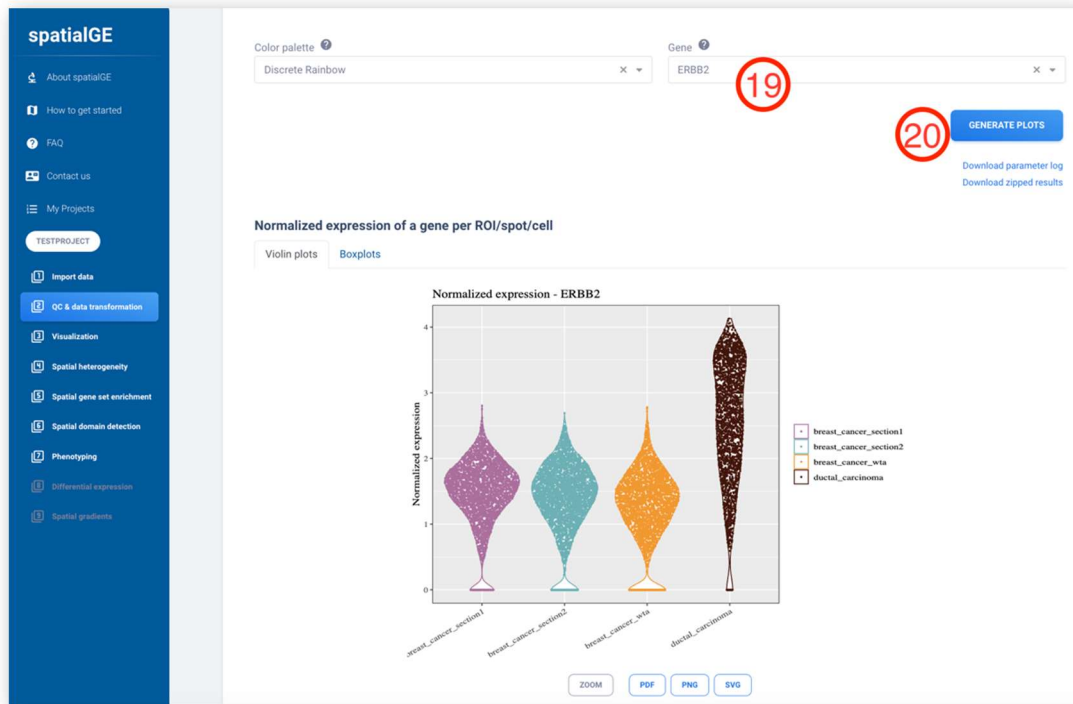
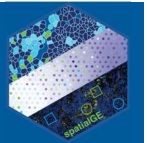
☐ Use SCTransform

NORMALIZE DATA

After the normalization procedure is completed, plots showing the count distribution by sample are displayed at the bottom of the page. These plots aim to facilitate the user's observation of the effects of normalization. Plots can be saved (**PDF/PNG/SVG** buttons, orange arrow), allowing the user to re-run the normalization with another method and decide which one provides the best results.



Scrolling down, users can find options to plot normalized counts per spot for specific genes (Count distribution by gene). The **Gene** text box is used to query for gene names to display violin plots or boxplots. The query is case-sensitive, and attention should be paid to the organism being studied (i.e., human genes are generally denoted by all uppercase letters; mouse genes are generally denoted with first letter uppercase). The plots can also be customized by selecting a **Color palette**. In the **Gene** text box, please type "ERBB2" and select it from the gene from the list. Then click **GENERATE PLOTS**. Each point in this plot represents a spot.

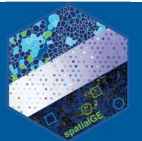


Next, please click on the **Pseudo-bulk analysis** tab.

Pseudo-bulk analysis

The **Pseudo-bulk analysis** tab provides functionality to assess sample-to-sample similarity. Pseudo-bulk counts are generated by aggregating the spot/cell counts within each sample, mimicking a “bulk” sample. The consequence is that spatial information is disregarded, and hence this analysis is only intended as a data QC method. The pseudo-bulk counts are used for dimension reduction via PCA or UMAP.

Once in the **Pseudo-bulk analysis** tab, start by selecting the **Number of most variable genes to perform dimensionality reduction (PCA/UMAP)**. Values between 3000 and 5000 (though not limited to this value), can be enough to visually assess sample similarity. Here, 3000 genes will be used. Now, please click **CALCULATE PCA AND UMAP**.



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Pseudo-bulk analysis explores sample-level gene expression. Pseudo-bulk analysis combines gene counts across all spots and produce pseudo-RNAseq samples. This analysis allows exploration of sample-level transcriptional patterns. Does not require execution of the Filter Data or Normalize Data sections.

- Quilt plot:** Generates spatial (quilt) plots showing spot/cell-level metadata. Users can assess QC metrics such as gene counts per spot or total counts per spot and detect potential technical artifacts.

Original summary Filter data Normalize data **Pseudo-bulk analysis** Quilt plot

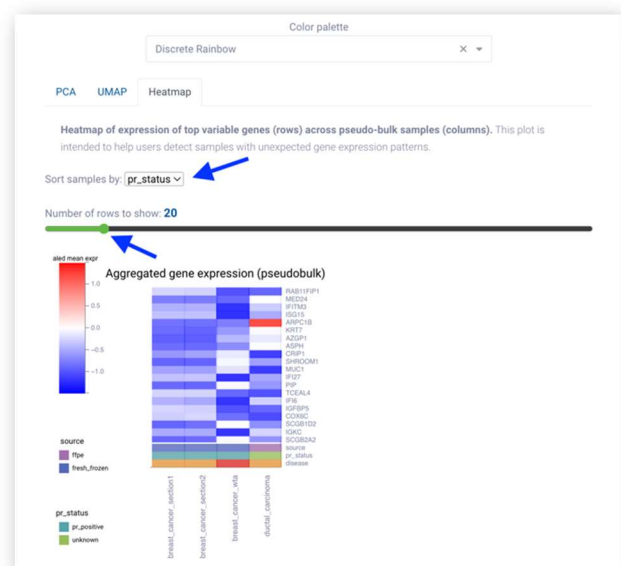
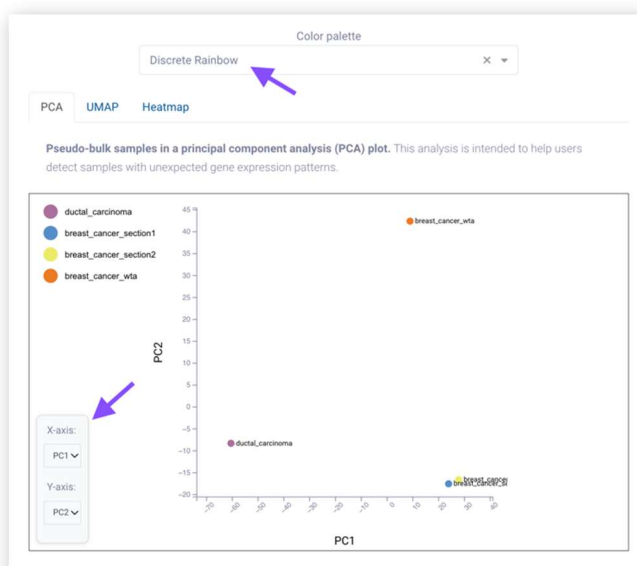
Dimensionality reduction of pseudo-bulk counts (PCA and UMAP)

The PCA and UMAP plots generated by this module result from calculating the average expression of genes across spots within each sample. This pseudo-bulk analysis does not incorporate any spatial component of the data. The PCA/UMAP is calculated with a number of user-selected most variable genes based on standard deviation. **Note:** Pseudo-bulk analysis can only be performed on projects with four samples or more.

Number of most variable genes to perform dimensionality reduction (PCA/UMAP):

CALCULATE PCA AND UMAP

The **CALCULATE PCA AND UMAP** button performs the aggregation of counts (pseudo-bulk counts) and the calculation of principal components (PCA) and UMAP embeddings. Once finished, three tabs become available: **PCA**, **UMAP**, and **Heatmap**. Color palettes can be changed for the **PCA** and **UMAP**, and multiple combinations of PCs can be compared (purple arrows). In the **Heatmap** tab, the **Number of genes to show** slider controls the number of heatmap rows to present (blue arrow), with each row displaying the scaled pseudo-bulk expression of a gene for all samples. Genes in the heatmap are chosen based on the coefficient of variation. The drop-down **Sort samples by** specifies how the samples (columns) are sorted in the heatmap. The variables in **Sort samples by** are directly extracted from the sample metadata entered during [data import](#).





Quilt plot

By clicking on the **quilt plot** tab, the user can generate plots to observe the variation in gene counts at each ROI/spot/cell, organized spatially. The quilt plots are a quick way to identify areas with higher counts (possibly higher cellular density) or, conversely, areas with low counts (possibly low cellular density or necrotic). The **Color palette** drop-down allows you to change the color of the ROIs/spots/cells in the quilt plot. Since the spots/cells in the quilt plot present quantitative data (total counts or total genes), the diverging and sequential palettes are recommended (see the [Khroma](#) and [RColorBrewer](#) documentation for more information on those types of palettes). For this example, please select the “Sunset” palette. Then, select “total_counts” from the **Variable** drop-down. Next, please select the “ductal_carcinoma” and “breast_cancer_section1” samples from the drop-downs “**First sample**” and “**Second sample**” respectively. Finally, click the “**GENERATE PLOTS**” button.

• **Filter data:** Remove ROIs/spots/cells or genes from the data set. The filter can be executed on all samples or a subset. Internally, spatialGE executes the filter in the following order: 1. Samples specified in “Select samples to apply this filter”, 2. Genes specified by name in “Filter genes”, 3. ROIs/spots/cells and/or genes based on min and max counts.

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Quilt Plot

Select two samples to compare the distribution of counts, genes, or any spot-level data in the spatial context.

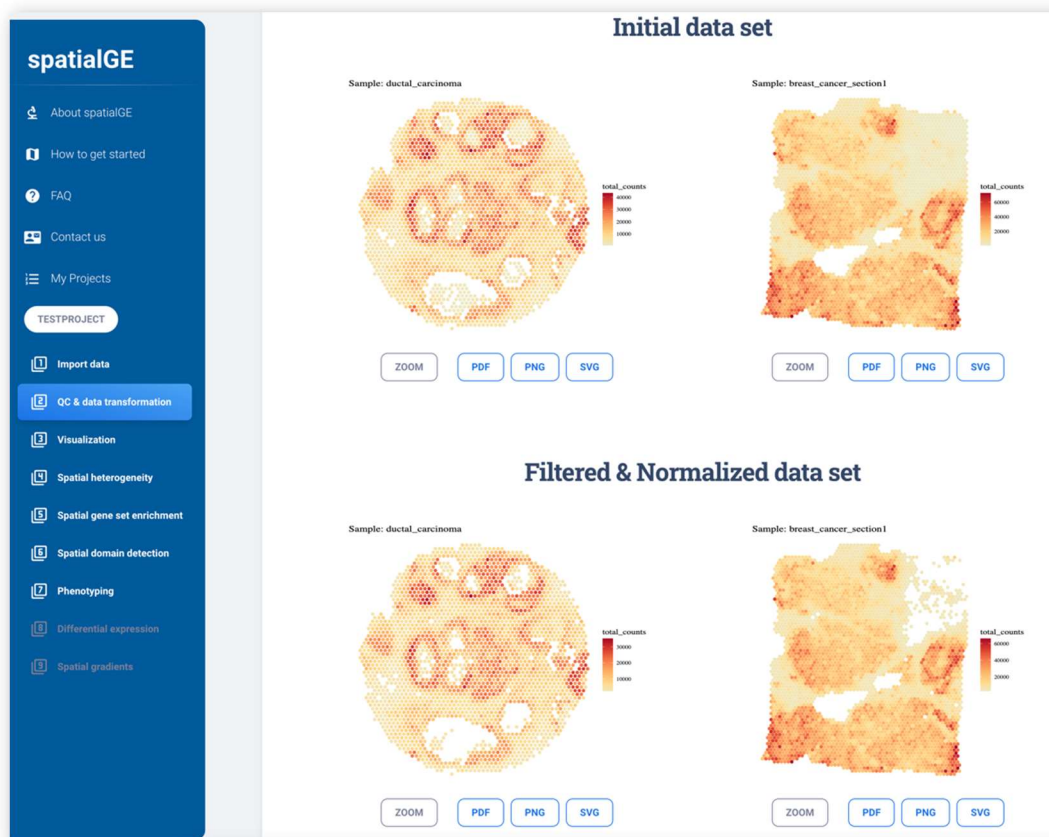
Color palette [?] Variable [?]

24 Sunset × 25 total_counts ×

First sample: [?] Second sample: [?]

26 ductal_carcinoma 27 breast_cancer_section1

GENERATE PLOTS



The resulting quilt plots show the tissue regions with high counts. If filtering and normalization have been performed, spatialGE produces before-and-after plots. Here, the user can see that some of the areas with low counts were removed after filtering. These figures can help the user decide if filtering has been too stringent or if different filtering options need to be applied.

Note: Currently, spatialGE only generates quilt plots for the total number of counts per spot and the total number of genes per spot (i.e., genes with one count or more per spot). Future versions of spatialGE will implement other QC metrics such as the percentage of mitochondrial genes.